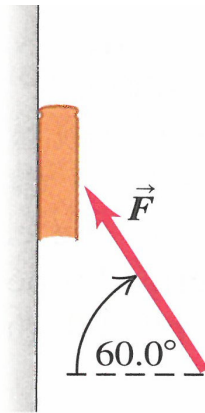
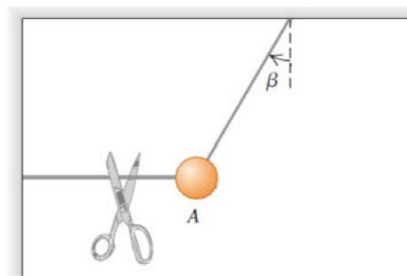


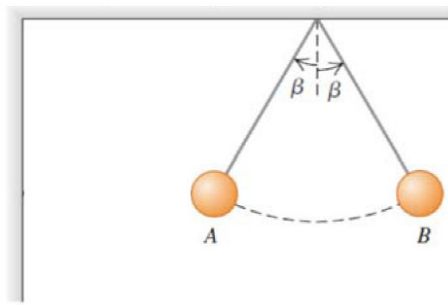
1. You place a book of mass 5.00 kg against a vertical wall, and apply a constant force F to the book, where $F = 96\text{ N}$. The force is applied at an angle of 60° from the horizontal as shown in the figure. The coefficient of kinetic friction between the book and the wall is 0.30 . If the book is initially at rest, what is its speed after it has traveled 0.40 m up the wall? (1.78 m/s)



2. A ball is held at rest at position A by two light strings, as shown in the following diagram.

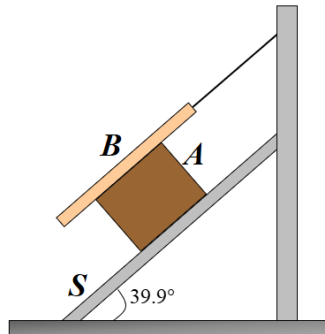


The horizontal string is subsequently cut, and the ball starts swinging as a pendulum. Position B as shown in the following diagram is the farthest to the right that the ball can go as it swings back and forth.



What is the ratio of the tension in the supporting string at **B** to its value at **A** **before** the string is cut?

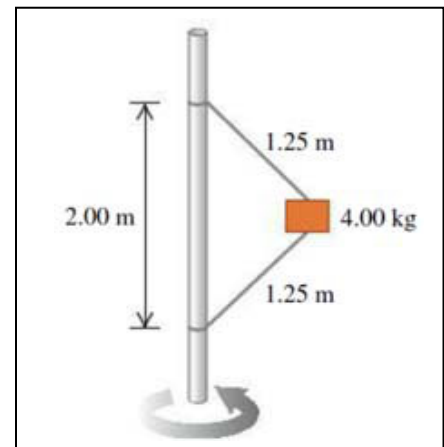
3. Block **A**, with weight $3w$, slides down an inclined plane **S** of slope angle 39.9° at a constant speed while plank **B**, with weight w , rests on top of **A**. The plank is attached by an inextensible cord to the wall as shown in the following diagram.



If the coefficient of kinetic friction is the same between **A** and **B**, and between **S** and **A**, determine its value. (0.5)

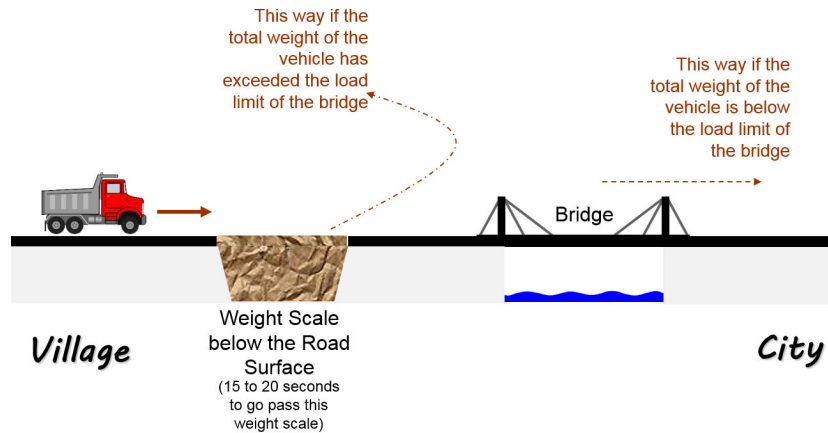
4. A 4.00 kg block is attached to a vertical rod by means of two strings. When the system rotates about the axis of the rod, the strings are extended as shown and the tension in the upper string is 80.0 N.

- What is the tension in the lower cord? (31 N)
- How many revolutions per minute does the system make? (44.99 rev/minute)
- Find the number of revolutions per minute at which the lower cord just goes slack. (29.89 rev/minute)
- Explain what happens if the number of revolutions per minute is less than that in part (iii).



5. To go on a bridge linking a village and a city, a sealed container truck will first have to go on a vehicle weight scale to check if the vehicle is within the load limit of the bridge. Suppose the time required for a vehicle to go pass the scale installed below the road surface in front of the bridge is 15 to 20 seconds (refer to figure). A truck driver wanted to smuggle 5 refugees from the village to the city, but the truck with the driver

and only one refugee had already exceeded the load limit. However, without the refugees the truck (with the driver) will be still within the load limit. So the driver came up with an interesting idea, that, when the truck was approaching the scale, he would instruct all refugees to jump. These refugees faithfully obeyed and they jumped at the same time, and the truck immediately went on the weight scale, indicating that the total weight of the truck was within the load limit. The driver was very happy, thinking that the smuggling of the 5 refugees would be successful.

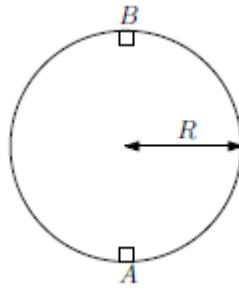


But, the refugees were not able to remain in the air forever and they fell on the floor after 0.8 second. Subsequently, the scale indicated that the total weight of the truck had exceeded the load limit of the bridge. So the driver had no choice but to drive the truck back to the village (refer to figure). Subsequently he and the 5 refugees were sentenced to prison for 20 days.

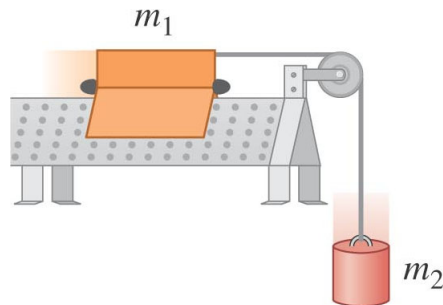


Thinking that the geese are able to hover in the air and will not touch the floor, the driver wants to smuggle geese after he is released from prison. Will the second attempt be successful? Why?

6. A small block with mass 0.0350 kg slides in a vertical circle of radius $r = 0.400 \text{ m}$ on the inside of a circular track. During one of the revolutions of the block, when the block is at the bottom of its path, point A , the normal force exerted on the block by the track has magnitude 4.05 N . In this same revolution, when the block reaches the top of its path, point B , the normal force exerted on the block has magnitude 0.690 N . How much work is done on the block by friction during the motion of the block from point A to point B ? (0.26 J)



7. If a fish is attached to a vertical spring and slowly lowered to its equilibrium position, it is found to stretch the spring by an amount d . If the same fish is attached to the end of the unstretched spring and then allowed to fall from rest, through what maximum distance does it stretch this spring?
8. A 2.90 kg fish is attached to the lower end of a vertical spring that has negligible mass and force constant 865 N/m. The spring initially is neither stretched nor compressed. The fish is then released from rest.
- (i) What is its speed after it has descended 0.0490 m from its initial position? (0.49 m/s)
- (ii) What is the maximum speed of the fish as it descends? (0.57 m/s)
9. An engine of mass 20 tons is attached to ten carriages each of mass 30 tons. The train is moving along level track at 30 km/h and is working at a rate of 400 kW. The resistance to motion are 1500 N for the engine, and 600 N for each carriage. Find the acceleration of the train and the tension in the coupling between the engine and the first carriage. (1 ton = 1000 kg.) (0.1266 m/s², 43968.75 N)
10. The figure shows an air-track glider with mass m_1 moving on a level and frictionless air track in the physics lab.

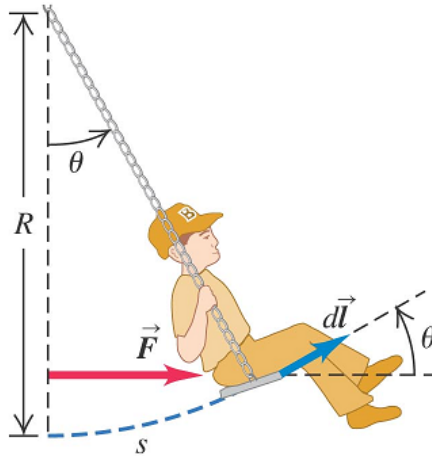


The glider is connected to a lab weight with mass m_2 by a light, flexible, non-stretching string that passes over a stationary, frictionless pulley. Find the acceleration a of each body and the tension T in the string.

11. You throw a 0.145 kg baseball straight up, giving it an initial velocity of magnitude 20.0 m/s. Suppose your hand moves upward by 0.500 m while you are throwing the ball. Ignore resistance. Find

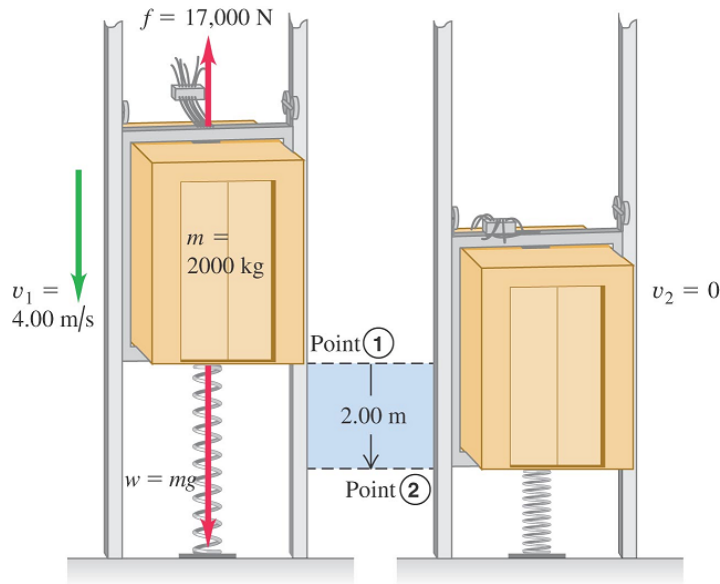
- (i) how high h the ball goes above the point where it leaves your hand. (20.4 m)
- (ii) the speed v of the ball at a point 15.0 m above the point where it leaves your hand. (10.3 m/s)
- (iii) the magnitude F of the force (assumed constant) that your hand exerts on the ball. (59.42 N)

12. At a family picnic you are appointed to push your cousin Bobby in a swing.



His weight is w , the length of the chain is R , and you push Bobby until the chains make an angle θ_0 with the vertical. To do this you exert a varying horizontal force F that starts at zero and gradually increases just enough that Bobby and the swing move very slowly and remain very nearly in equilibrium throughout the process. Neglect the weight of the chains and seat.

- (i) What is the work done W_T by the tension T in the chains?
 - (ii) What is the total work done W_{total} on Bobby by all forces?
 - (iii) What is the work W_F you do by exerting the force F ?
13. A 2000 kg elevator with broken cables in a test rig is falling at 4.00 m/s when it contacts a cushioning spring at the bottom of the shaft. The spring is intended to stop the elevator, compressing 2.00 m as it does so. During the motion a safety clamp applies a constant 17000 N frictional force to the elevator. What is the necessary force constant k for the spring? (10600 N/m)



14. You are an industrial engineer with a shipping company. As part of the package-handling system, a small box with mass 1.60 kg is placed against a light spring that is compressed 0.280 m . The spring, whose other end is attached to a wall, has force constant $k = 45.0 \text{ N/m}$. The spring and box are released from rest, and the box travels along a horizontal surface for which the coefficient of kinetic friction with the box is $\mu_k = 0.300$. When the box has traveled 0.280 m and the spring has reached its equilibrium length, the box loses contact with the spring.
- What is the speed of the box at the instant when it leaves the spring?
(0.75 m/s)
 - What is the maximum speed of the box during the restoration of spring back to its natural length? (0.93 m/s)
 - Why are the two speeds in (i) and (ii) not equal?
15. A pump working at 100 kW pumps water horizontally through a pipe of radius 30 cm . Find the volume of water pumped per second assuming that the water is initially at rest.
(2.52 m³)

- End -

Formula List

Logarithm:

$$\log a + \log b = \log(a \times b)$$

$$\log a - \log b = \log\left(\frac{a}{b}\right)$$

$$(\log x) > 0, \text{ if } x > 1.$$

$$(\log 1) = 0.$$

$$(\log x) < 0, \text{ if } 0 < x < 1.$$

$$(\log x) \text{ is undefined if } x < 0.$$

Trigonometry:

$$\sin = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin}{\cos}$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Differentiation:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(e^{f(x)}) = e^{f(x)} \times \frac{d}{dx}(f(x))$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(f(x) \times g(x)) = f(x) \times \left(\frac{d}{dx}g(x)\right) + \left(\frac{d}{dx}f(x)\right) \times g(x)$$

Integration:

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c$$

Vector:

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$

$$\vec{A} \bullet \vec{B} = |\vec{A}| \times |\vec{B}| \times \cos \theta$$

Linear kinematic equations:

$$v_f = v_i + at$$

$$\Delta x = v_i t + \frac{1}{2} at^2$$

$$\Delta x = \frac{(v_i + v_f)}{2} \times t$$

$$v_f^2 = v_i^2 + 2a\Delta x$$

Simple harmonic motion:

$$x(t) = r_0 \cos(\theta_0 + \omega_0 t)$$

$$v_x(t) = -r_0 \omega_0 \sin(\theta_0 + \omega_0 t)$$

$$a_x(t) = -r_0 \omega_0^2 \cos(\theta_0 + \omega_0 t) = -\omega_0^2 x(t)$$

$$\omega^2 = \frac{k}{m}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$v_{\max} = \sqrt{\frac{k}{m}} A$$

$$a_{\max} = \frac{k}{m} A$$

Retarding force:

$$R = -bv$$

Projectile motion:

$$y = -\frac{g}{2v_{0x}^2}(x - x_0)^2 + \frac{v_{0y}}{v_{0x}}(x - x_0) + y_0, \text{ or}$$

$$y = x \tan \theta - \frac{gx^2}{2v_0^2}(1 + \tan^2 \theta)$$

Uniform circular motion:

$$\theta = \frac{\text{arc length}}{\text{radius}} = \frac{s}{r}$$

$$\omega = \frac{d\theta}{dt} = \frac{2\pi}{T}$$

$$v = r\omega$$

$$a_c = r\omega^2 = \frac{v^2}{r}$$

Non-uniform circular motion:

$$a_t = \frac{dv}{dt}$$

$$a = \sqrt{a_c^2 + a_t^2}$$

Relative velocity:

$$\vec{V}_{P/A} = \vec{V}_{P/B} + \vec{V}_{B/A}$$

Newton's second law:

$$\sum \vec{F} = m\vec{a}$$

Newton's third law:

$$\vec{F}_{12} = -\vec{F}_{21}, \text{ and the two forces} \\ \text{act on the opposite body.}$$

Frictional force:

$$f_s \leq \mu_s n, \text{ and impending motion} \\ \text{occurs when } f_s = \mu_s n.$$

$$f_k = \mu_k n$$

Translational kinetic energy:

$$KE_{\text{translational}} = \frac{1}{2}mv^2$$

Gravitational potential energy:

$$\Delta PE = mg(y_2 - y_1) \\ = mgh$$

Spring

$$F_{\text{spring}} = -kx$$

$$\text{Elastic } PE_{\text{spring}} = \frac{1}{2}kx^2$$

Work-energy theorem:

$$W = \Delta KE = KE_f - KE_i$$

Work done (constant force):

$$W = \vec{F} \bullet \Delta \vec{r} = F \Delta r \cos \theta$$

Work done (varying force):

$$W = \int_{x_i}^{x_f} \vec{F} \bullet d\vec{r}$$

Power:

$$P = \frac{dW}{dt} = F \times v$$

Linear momentum:

(also valid for y and z)

$$p_x = mv_x$$

$$F_x = \frac{dp_x}{dt}$$

$$F_{\text{average, x}} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} F_x(t) dt$$

Conservation of linear momentum:

$$\vec{p}_{1,i} + \vec{p}_{2,i} = \vec{p}_{1,f} + \vec{p}_{2,f}$$

Elastic collision:

$$v_{A1} - v_{B1} = -(v_{A2} - v_{B2})$$

$$v_{A2} = \left(\frac{m_A - m_B}{m_A + m_B} \right) v_{A1} + \left(\frac{2m_B}{m_A + m_B} \right) v_{B1}$$

$$v_{B2} = \left(\frac{m_B - m_A}{m_A + m_B} \right) v_{B1} + \left(\frac{2m_A}{m_A + m_B} \right) v_{A1}$$

Center of mass

(also valid for y and z)

$$x_{\text{cm}} = \frac{1}{M} \sum m x, \text{ where } M = \sum m, \text{ or}$$

$$= \frac{1}{M} \int x \, dm$$

$$v_{\text{cm},x} = \frac{1}{M} \sum m v_x$$

$$a_{\text{cm},x} = \frac{1}{M} \sum m a_x$$

Simple Harmonic Motion:

$$a_x = -\omega^2 x$$

$$\omega^2 = \frac{k}{m}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$v_{\text{max}} = \sqrt{\frac{k}{m}} A$$

$$a_{\text{max}} = \frac{k}{m} A$$

Retarding force:

$$F = -b v$$

Rotation

$$\theta = \frac{s}{r}$$

$$\omega = \frac{d\theta}{dt}$$

$$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

$$v_{\text{tangential}} = r\omega$$

$$a_{\text{centripetal}} = r\omega^2 = \frac{v^2}{r}$$

$$a_{\text{tangential}} = r\alpha$$

Rotational kinematic equations:

$$\omega_f = \omega_i + \alpha t$$

$$\Delta\theta = \omega_i t + \frac{1}{2} \alpha t^2$$

$$\Delta\theta = \frac{(\omega_i + \omega_f)}{2} \times t$$

$$\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$$

Moment of inertia:

$$I = \sum m r^2$$

$$I = \int r^2 \, dm$$

Parallel axis theorem

$$I = I_{\text{cm}} + M d^2$$

Rotational kinetic energy:

$$KE_{\text{rotational}} = \frac{1}{2} I \omega^2$$

Moment of inertia:

$$I = \sum m_i r_i^2$$

$$I = \int r^2 \, dm$$

Moment of inertia of hoop or thin cylindrical shell:

$$I_{\text{CM}} = MR^2$$

Moment of inertia of hollow cylinder:

$$I_{\text{CM}} = \frac{1}{2} M (R_1^2 + R_2^2)$$

Moment of inertia of solid cylinder:

$$I_{\text{CM}} = \frac{1}{2} MR^2$$

Moment of inertia of rectangular plate:

$$I_{\text{CM}} = \frac{1}{12} M (a^2 + b^2)$$

Moment of inertia of thin rod about its CM:

$$I_{CM} = \frac{1}{12} ML^2$$

Moment of inertia of thin rod about its end:

$$I = \frac{ML^2}{3}$$

Moment of inertia of solid sphere:

$$I_{CM} = \frac{2}{5} MR^2$$

Moment of inertia of spherical shell:

$$I_{CM} = \frac{2}{3} MR^2$$

Torque and Moment of force:

$$\tau = F \times r \times \sin \theta$$

$$\sum \tau = I\alpha$$

Work due to rotation:

$$\Delta W = \tau \times \Delta \theta$$

$$W = \int_{\theta_1}^{\theta_2} \tau d\theta$$

Power due to rotation:

$$P = \tau \times \omega$$

Pure rolling:

$$v_{cm} = r\omega$$

$$a_{cm} = r\alpha$$

$$KE_{\text{total}} = \frac{1}{2} Mv_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$$

Conservation of Angular momentum:

$$\underbrace{\sum I\omega}_{\text{Before collision}} = \underbrace{\sum I\omega}_{\text{After collision}}$$

Avogadro's number:

$$N_A \approx 6.022 \times 10^{23} \text{ particles/mole}$$

Ideal gas constant:

$$R \approx 8.314 \text{ J mole}^{-1} \text{ K}^{-1}$$

Boltzmann constant:

$$k = \frac{R}{N_A} \approx 1.381 \times 10^{-23} \text{ J/K}$$

Ideal gas equation:

$$pV = nRT$$

Speed of a molecule

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

Mole

$$n = \frac{N}{N_A}$$

$$n = \frac{m}{M}$$

Absolute temperature

$$T = T_{\text{Celsius}} + 273.15$$

Specific heat c and molar heat C

$$Q = Mc(T_2 - T_1)$$

$$= nC(T_2 - T_1)$$

Latent heat L

$$Q = L \times M$$

First law of thermodynamics

$$\Delta U = Q - W$$

Internal energy for N molecules

$$\text{monatomic: } U = N \times \frac{3}{2} kT$$

$$\text{diatomic: } U = N \times \frac{5}{2} kT$$

Relationship between C_p and C_v for an ideal Gas

$$C_p - C_v = R$$

Isochoric process

$$\text{monatomic: } C_V = \frac{3}{2}R$$

$$\text{diatomic: } C_V = \frac{5}{2}R$$

C_p is undefined for isochoric process.

Isobaric process

$$\text{monatomic: } C_p = \frac{5}{2}R$$

$$\text{diatomic: } C_p = \frac{7}{2}R$$

C_V is undefined for isobaric process.

Isothermal process

$$W = nRT \ln \left(\frac{V_f}{V_i} \right)$$

Adiabatic process

$$\gamma = \frac{C_p}{C_V}$$

$$\text{monatomic: } \gamma = \frac{5}{3}$$

$$\text{diatomic: } \gamma = \frac{7}{5}$$

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$P_1^{1-\gamma} T_1^\gamma = P_2^{1-\gamma} T_2^\gamma$$

Thermal efficiency of a heat engine

$$e = 1 - \frac{|Q_C|}{|Q_H|}$$

Coefficient of performance of a heat pump for cooling

$$\text{COP} = \frac{|Q_C|}{|Q_H| - |Q_C|}$$

Entropy

$$\Delta S = \int_{\text{initial state}}^{\text{final state}} \frac{dQ}{T} \geq 0$$

Coulomb's Law:

$$F = \frac{kq_1 q_2}{r^2}$$

where

$$k = 8.9875 \times 10^9 \text{ Nm}^2/\text{C}^2$$

Electric field:

$$E = \frac{F}{q_0}$$

Electric field for point charge:

$$E = \frac{kq}{r^2}$$

Electric field at a distance along central axis of charged ring:

$$E = \frac{kQx}{(x^2 + a^2)^{3/2}}$$

Electric potential:

$$V = \frac{U}{q_0}$$

Electric potential in uniform field:

$$\Delta V = -Ed$$

Electric potential for point charge:

$$V = \frac{kq}{r}$$

Electric potential at a distance along central axis of charged ring:

$$V = \frac{kQ}{\sqrt{x^2 + a^2}}$$

Electric potential energy (for multiple point charges):

$$U = k \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right)$$

Electric current:

$$I = \frac{dQ}{dt}$$
$$I_{av} = nqv_d A$$

Ohm's Law:

$$I = \frac{\Delta V}{R}$$

Resistance of ohmic conductor:

$$R = \rho \frac{L}{A}$$

Resistivity:

$$\rho = \rho_0 [1 + \alpha(T - T_0)]$$

Electric power:

$$P = I\Delta V = I^2 R = \frac{\Delta V^2}{R}$$

Electric energy:

$$U = P\Delta t$$

Terminal voltage:

$$\Delta V = \mathcal{E} - Ir$$

Resistors in series:

$$R_{eq} = R_1 + R_2 + R_3 + \dots$$

Resistors in parallel:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

Kirchhoff's Junction Rule:

$$\sum I_{in} = \sum I_{out}$$

Kirchhoff's Loop Rule:

$$\sum \Delta V = 0$$

Magnetic force on a moving charge:

$$F_B = qvB \sin \theta$$

Magnetic force on a current carrying conductor:

$$F = IlB \sin \theta$$

Radius of moving charged particle in uniform magnetic field:

$$r = \frac{mv}{qB}$$

Angular speed:

$$\omega = \frac{qB}{m}$$

Period:

$$T = \frac{2\pi m}{qB}$$

Velocity selector:

$$v = \frac{E}{B}$$

Magnetic field due to long straight conductor:

$$B = \frac{\mu_0 i}{2\pi a} \text{ where}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$$

Magnetic field at center of circular wire loop:

$$B = \frac{\mu_0 I}{2R}$$

Magnetic field at a distance along central axis of current-carrying ring:

$$B = \frac{\mu_0 IR^2}{2(x^2 + R^2)^{3/2}}$$

Magnetic force between parallel conductors:

$$F = \frac{\mu_0 I_1 I_2 l}{2\pi a}$$

Magnetic field inside toroid:

$$B = \frac{\mu_0 NI}{2\pi r}$$

Magnetic field inside solenoid:

$$B = \mu_0 nI$$

Magnetic flux:

$$\Phi_B = BA \cos \theta$$

Faraday's Law:

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}$$

Motional emf:

$$\mathcal{E} = \frac{Blv}{R}$$

Power input:

$$P = F_{\text{app}} v = I l B v$$

Mass

Electron: 9.1094×10^{-31} kg

Proton: 1.6726×10^{-27} kg

Neutron: 1.6749×10^{-27} kg

Charge

Electron: -1.6022×10^{-19} C

Proton: $+1.6022 \times 10^{-19}$ C

Neutron: 0

